

# Video Tutor Agent

Paste a YouTube link, then **talk** to a tutor that answers only from the video, in English, Hindi, Hinglish or other Indian languages. Built entirely on **Sarvam**.

Code: [github.com/tjay27/video-tutor-agent](https://github.com/tjay27/video-tutor-agent)

## Scope

- **In:** YouTube link → transcript; a **hands-free voice conversation** you can interrupt; answers grounded in the video with **timestamp citations**; Indian-language and code-mixed speech; web UI.
- **Out (for now):** videos over 60 min / RAG; what is shown on screen (slides, diagrams); accounts, database, hosting.

## Architecture

- **Transcript builder:** yt-dlp (audio + captions) → **Sarvam saaras:v4** batch STT (codemix, key terms from channel and tags) → sentence timestamps snapped to caption timing, gaps filled from captions → cached. **~40 s** for a 9-min video.
- **Voice session** (WebSockets): mic → **saaras:v3-realtime** STT → **turn analyzer** (small LLM call: question / unfinished / stop / not for the tutor) → **sarvam-105b** with the full transcript, streamed → **bulbul:v3** streaming TTS, sentence by sentence. **~1.2–1.5 s** from the student going quiet to hearing the tutor.
- **Web UI** (plain HTML/JS): YouTube player, searchable transcript, chat with timestamp chips, hold-to-talk fallback.
- **No RAG, on purpose:** full transcript in the prompt. Measured time to first word: 9 min **0.4 s**, 1 h **~1 s**, 2 h **1.9 s**. RAG only above ~1 hour.

## Challenges & fixes

- **Slow answers (14 s, sometimes empty):** LLM reasoning mode → turned off: **0.5–0.9 s**.
- **Poor Hinglish transcripts:** captions wrote English in Hindi script; **saaras:v3 silently dropped 2 minutes** → saaras:v4 codemix + completeness check.
- **Cut off mid-thought:** fixed silence timers → the turn analyzer judges whether the thought is **finished by meaning**; pauses are merged into one question.
- **Background chatter and music:** got answered, or froze the tutor → speech not for the tutor is **ignored**; the tutor **pauses** when anyone speaks and **resumes from the same spot** if it wasn't for it.
- **Interruptions:** volume ducking sounded unnatural → clean stop + a natural filler ("Sure.") before the new answer; "ruko / stop" → "Okay." and silence; "hmm" never interrupts.
- **Real questions ignored as the tutor's own voice:** crude echo check → echo = the tutor's words coming back **word for word**, only while it is playing.
- **Wrong reply language:** STT's language tag gets "stuck" → reply language judged from the words' script; an explicit "talk in English" is remembered.
- **Outside facts, bad timestamps:** strict "video only" rules + a **citation check** that snaps every timestamp to a real transcript line.
- **An answer went silent:** one digits-only chunk made TTS fail the whole answer → such chunks are skipped; one bad sentence no longer stops the rest.

## Results

Turn analyzer **20/21** on labelled speech (~200 ms). End-to-end voice scenarios **8/8**. Every issue, cause and fix is logged in CHALLENGES.md.